

UNIT-7: MATRICES & DETERMINANTS

Comprehensive Practice Workbook: 25 Fully Solved Strategic Problems

Exhaustive Training Sheet • Complete Element Algebra Solutions Without LaTeX Syntax

Section 1: Matrix Addition, Subtraction, and Linear Scaling

PROBLEM 1: ADDITION OF 2X2 MATRICES

Given matrices A and B , find $A + B$:

$$A = \begin{vmatrix} 4 & -1 \\ 3 & 7 \end{vmatrix}, \quad B = \begin{vmatrix} 2 & 5 \\ -6 & 1 \end{vmatrix}$$

SOLUTION:

Add the corresponding entries from each position:

Row 1: $[4 + 2, -1 + 5] = [6, 4]$

Row 2: $[3 + (-6), 7 + 1] = [-3, 8]$

$$A + B = \begin{vmatrix} 6 & 4 \\ -3 & 8 \end{vmatrix}$$

PROBLEM 2: SUBTRACTION OF 2X2 MATRICES

Using matrices **A** and **B** from Problem 1, calculate **A - B**.

SOLUTION:

Subtract each entry of matrix **B** from the corresponding entry in matrix **A**:

Row 1: $[4 - 2, -1 - 5] = [2, -6]$

Row 2: $[3 - (-6), 7 - 1] = [9, 6]$

$$\mathbf{A} - \mathbf{B} = \begin{vmatrix} 2 & -6 \\ 9 & 6 \end{vmatrix}$$

PROBLEM 3: SCALAR SCALING OF A 2X3 MATRIX

Multiply the following matrix **M** by the scalar factor $k = -3$:

$$\mathbf{M} = \begin{vmatrix} 1 & -4 & 0 \\ 2 & 5 & -2 \end{vmatrix}$$

SOLUTION:

Multiply every entry inside the matrix by -3:

Row 1: $[-3(1), -3(-4), -3(0)] = [-3, 12, 0]$

Row 2: $[-3(2), -3(5), -3(-2)] = [-6, -15, 6]$

$$-3\mathbf{M} = \begin{vmatrix} -3 & 12 & 0 \\ -6 & -15 & 6 \end{vmatrix}$$

PROBLEM 4: LINEAR COMBINATION OF 3X3 FORMATS

Find the matrix combination $2\mathbf{A} + 3\mathbf{B}$ given:

$$\mathbf{A} = \begin{vmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 2 & 1 & 0 \end{vmatrix}, \quad \mathbf{B} = \begin{vmatrix} 0 & 2 & 1 \\ 1 & -2 & 3 \\ 0 & 4 & 1 \end{vmatrix}$$

SOLUTION:

First, scale both matrices individually:

$2\mathbf{A} =$

$$\begin{vmatrix} 2 & 0 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 0 \end{vmatrix}, \quad 3\mathbf{B} = \begin{vmatrix} 0 & 6 & 3 \\ 3 & -6 & 9 \\ 0 & 12 & 3 \end{vmatrix}$$

Next, sum the two scaled matrices together:

$$2\mathbf{A} + 3\mathbf{B} = \begin{vmatrix} 2 & 6 & 7 \\ 1 & 0 & 11 \\ 4 & 14 & 3 \end{vmatrix}$$

PROBLEM 5: SOLVING A LINEAR MATRIX EQUATION

Find the matrix X that satisfies the equation $2A + X = B$, where:

$$A = \begin{vmatrix} 1 & 3 \\ -2 & 4 \end{vmatrix}, \quad B = \begin{vmatrix} 5 & 1 \\ 0 & 6 \end{vmatrix}$$

SOLUTION:

Rearrange the equation to isolate X : $X = B - 2A$.

Calculate $2A$:

$2A =$

$$\begin{vmatrix} 2 & 6 \\ -4 & 8 \end{vmatrix}$$

Now subtract $2A$ from B :

$$X = \begin{vmatrix} 5 - 2 & 1 - 6 \\ 0 - (-4) & 6 - 8 \end{vmatrix} = \begin{vmatrix} 3 & -5 \\ 4 & -2 \end{vmatrix}$$

Section 2: Matrix Multiplication Mechanics

PROBLEM 6: MULTIPLICATION OF 2X2 SQUARE MATRICES

Calculate the product matrix $\mathbf{AB}$ for:

$$\mathbf{A} = \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix}, \quad \mathbf{B} = \begin{vmatrix} 5 & -1 \\ 0 & 2 \end{vmatrix}$$

SOLUTION:

Apply the row-by-column dot product algorithm:

$$c_{11} = (2 \times 5) + (3 \times 0) = 10 + 0 = 10$$

$$c_{12} = (2 \times -1) + (3 \times 2) = -2 + 6 = 4$$

$$c_{21} = (1 \times 5) + (4 \times 0) = 5 + 0 = 5$$

$$c_{22} = (1 \times -1) + (4 \times 2) = -1 + 8 = 7$$

$$\mathbf{AB} = \begin{vmatrix} 10 & 4 \\ 5 & 7 \end{vmatrix}$$

PROBLEM 7: VERIFYING NON-COMMUTATIVITY (BA PRODUCT)

Using the same matrices from Problem 6, calculate $\mathbf{BA}$ to demonstrate that matrix multiplication is non-commutative ($\mathbf{AB} \neq \mathbf{BA}$).

SOLUTION:

Perform the multiplication in reverse order ($\mathbf{B} \times \mathbf{A}$):

$$d_{11} = (5 \times 2) + (-1 \times 1) = 10 - 1 = 9$$

$$d_{12} = (5 \times 3) + (-1 \times 4) = 15 - 4 = 11$$

$$d_{21} = (0 \times 2) + (2 \times 1) = 0 + 2 = 2$$

$$d_{22} = (0 \times 3) + (2 \times 4) = 0 + 8 = 8$$

$$\mathbf{BA} = \begin{vmatrix} 9 & 11 \\ 2 & 8 \end{vmatrix}$$

Comparing the results confirms that $\mathbf{AB} \neq \mathbf{BA}$, as matrix $[10, 4; 5, 7]$ does not equal $[9, 11; 2, 8]$.

PROBLEM 8: ROW MATRIX MULTIPLIED BY A COLUMN MATRIX

Compute the product $R \times C$ for row matrix R and column matrix C :

$$R = \begin{vmatrix} 2 & -1 & 4 \end{vmatrix}, C = \begin{vmatrix} 3 \\ 5 \\ -2 \end{vmatrix}$$

SOLUTION:

Matrix R is 1×3 and matrix C is 3×1 . The product will be a 1×1 single scalar entry:

$$\text{Product} = (2 \times 3) + (-1 \times 5) + (4 \times -2) = 6 - 5 - 8 = -7$$

$$RC = \begin{vmatrix} -7 \end{vmatrix}$$

PROBLEM 9: COLUMN MATRIX MULTIPLIED BY A ROW MATRIX

Using matrices R and C from Problem 8, compute the product $C \times R$.

SOLUTION:

Multiplying a 3×1 matrix by a 1×3 matrix results in a larger 3×3 matrix. Each entry c_{ij} is computed by multiplying row i of matrix C by column j of matrix R :

Row 1: $[3(2), 3(-1), 3(4)] = [6, -3, 12]$

Row 2: $[5(2), 5(-1), 5(4)] = [10, -5, 20]$

Row 3: $[-2(2), -2(-1), -2(4)] = [-4, 2, -8]$

$$CR = \begin{bmatrix} 6 & -3 & 12 \\ 10 & -5 & 20 \\ -4 & 2 & -8 \end{bmatrix}$$

PROBLEM 10: MULTI-ROW/COLUMN RECTANGULAR ARRAY PRODUCT

Find the product $\mathbf{AB}$ for the following rectangular matrices:

$$\mathbf{A} = \begin{vmatrix} 1 & 2 \\ 3 & 0 \\ -1 & 4 \end{vmatrix}, \quad \mathbf{B} = \begin{vmatrix} 2 & 1 & 3 \\ -1 & 0 & 2 \end{vmatrix}$$

SOLUTION:

Matrix $\mathbf{A}$ is 3×2 and matrix $\mathbf{B}$ is 2×3 . The product $\mathbf{AB}$ will be a 3×3 square matrix:

- Row 1: $[(1 \times 2) + (2 \times -1), (1 \times 1) + (2 \times 0), (1 \times 3) + (2 \times 2)] = [0, 1, 7]$
- Row 2: $[(3 \times 2) + (0 \times -1), (3 \times 1) + (0 \times 0), (3 \times 3) + (0 \times 2)] = [6, 3, 9]$
- Row 3: $[(-1 \times 2) + (4 \times -1), (-1 \times 1) + (4 \times 0), (-1 \times 3) + (4 \times 2)] = [-6, -1, 5]$, wait: $-3 + 8 = 5$

Let's double check Row 3: $(-1 \times 2) + (4 \times -1) = -2 - 4 = -6$. Second entry: $(-1 \times 1) + (4 \times 0) = -1$. Third entry: $(-1 \times 3) + (4 \times 2) = -3 + 8 = 5$.

$$\mathbf{AB} = \begin{vmatrix} 0 & 1 & 7 \\ 6 & 3 & 9 \\ -6 & -1 & 5 \end{vmatrix}$$

Section 3: Determinants of Matrices

PROBLEM 11: DETERMINANT OF A 2×2 MATRIX

Calculate the determinant of matrix $\mathbf{P}$:

$$\mathbf{P} = \begin{vmatrix} 5 & 3 \\ 2 & 4 \end{vmatrix}$$

SOLUTION:

Use the standard 2×2 formula $\det(\mathbf{P}) = ad - bc$:

$$\det(\mathbf{P}) = (5 \times 4) - (3 \times 2) = 20 - 6 = 14$$

PROBLEM 12: DETERMINANT WITH NEGATIVE ENTRIES

Find the determinant of the following matrix Q :

$$Q = \begin{vmatrix} -2 & 7 \\ 3 & -5 \end{vmatrix}$$

SOLUTION:

$$\det(Q) = (-2 \times -5) - (7 \times 3) = 10 - 21 = -11$$

PROBLEM 13: DETERMINANT OF A 3X3 MATRIX VIA ROW EXPANSION

Evaluate the determinant of matrix A :

$$A = \begin{vmatrix} 1 & 2 & 3 \\ 0 & -4 & 1 \\ 2 & 3 & 0 \end{vmatrix}$$

SOLUTION:

Expand along the first row, applying the alternating sign pattern (+, -, +):

$$|A| = 1 \times [(-4 \times 0) - (1 \times 3)] - 2 \times [(0 \times 0) - (1 \times 2)] + 3 \times [(0 \times 3) - (-4 \times 2)]$$

$$|A| = 1 \times (0 - 3) - 2 \times (0 - 2) + 3 \times (0 + 8)$$

$$|A| = 1(-3) - 2(-2) + 3(8) = -3 + 4 + 24 = 25$$

PROBLEM 14: DETERMINANT EXPANSION USING A ROW WITH ZERO ENTRIES

Evaluate the determinant of matrix B by expanding along its second row to simplify the calculations:

$$B = \begin{vmatrix} 2 & 4 & 1 \\ 3 & 0 & 5 \\ 1 & 2 & 1 \end{vmatrix}$$

SOLUTION:

The alternating sign pattern for the second row is $(-, +, -)$. Expanding along row 2 yields:

$$|B| = -3 \times [(4 \times 1) - (1 \times 2)] + 0 \times [...] - 5 \times [(2 \times 2) - (4 \times 1)]$$

$$|B| = -3 \times (4 - 2) + 0 - 5 \times (4 - 4)$$

$$|B| = -3(2) + 0 - 5(0) = -6 + 0 - 0 = -6$$

Section 4: Properties of Determinants

PROBLEM 15: DEMONSTRATING THE PROPORTIONAL ROW PROPERTY

Prove that the determinant of the following matrix M is zero without performing a full expansion:

$$M = \begin{vmatrix} 1 & 3 & 2 \\ 4 & 12 & 8 \\ 5 & 1 & 7 \end{vmatrix}$$

SOLUTION:

Observe the relationship between the rows of the matrix: Row 2 is exactly 4 times Row 1 ($R_2 = 4 \times R_1$).

According to the proportionality property of determinants, if any two rows or columns are exact scalar multiples of each other, the total determinant is automatically zero. Therefore, $\det(M) = 0$.

PROBLEM 16: EVALUATING THE DETERMINANT OF AN UPPER TRIANGULAR MATRIX

Calculate the determinant of the following upper triangular matrix T :

$$T = \begin{vmatrix} 3 & 7 & -2 \\ 0 & 2 & 5 \\ 0 & 0 & -4 \end{vmatrix}$$

SOLUTION:

For any upper triangular, lower triangular, or diagonal matrix, the determinant equals the product of the elements along the main diagonal.

Multiply the diagonal entries together:

$$\det(T) = t_{11} \times t_{22} \times t_{33} = 3 \times 2 \times (-4) = -24$$

PROBLEM 17: SIGN REVERSAL PROPERTY FROM ROW INTERCHANGES

Given matrix A with $\det(A) = 15$, what is the determinant of matrix B formed by swapping Row 1 and Row 3 of matrix A ?

SOLUTION:

According to the sign reversal property of determinants, interchanging any two rows or columns switches the sign of the determinant.

Therefore, $\det(B) = -\det(A) = -15$.

PROBLEM 18: DETERMINANT OF A SCALED MATRIX SYSTEM

If A is a 3×3 square matrix with a determinant value of $\det(A) = -4$, find the determinant of the scaled matrix $3A$.

SOLUTION:

When an $n \times n$ matrix is multiplied by a scalar constant k , the determinant scales exponentially based on the matrix's order: $\det(kA) = k^n \times \det(A)$.

Here, the matrix order is $n = 3$ and the scalar is $k = 3$:

$$\det(3A) = 3^3 \times \det(A) = 27 \times (-4) = -108$$

Section 5: Adjoint Matrix Formulations

PROBLEM 19: FINDING THE ADJOINT OF A 2X2 MATRIX

Find the adjoint matrix $\text{adj}(A)$ for:

$$A = \begin{vmatrix} 6 & -2 \\ 4 & 5 \end{vmatrix}$$

SOLUTION:

To find the adjoint of a 2x2 matrix, swap the elements on the main diagonal and flip the signs of the elements on the opposite diagonal:

Swap diagonal elements (6 and 5) $\rightarrow a_{11} = 5, a_{22} = 6$.

Negate opposite entries (-2 and 4) $\rightarrow a_{12} = 2, a_{21} = -4$.

$$\text{adj}(A) = \begin{vmatrix} 5 & 2 \\ -4 & 6 \end{vmatrix}$$

PROBLEM 20: FINDING THE ADJOINT OF A 3X3 MATRIX

Construct the adjoint matrix $\text{adj}(M)$ for the matrix:

$$M = \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{vmatrix}$$

SOLUTION:

First, calculate the cofactors for all nine positions ($C_{ij} = (-1)^{i+j} * M_{ij}$):

- $C_{11} = +(1-0) = 1$; $C_{12} = -(2-0) = -2$; $C_{13} = +(6-0) = 6$
- $C_{21} = -(0-6) = 6$; $C_{22} = +(1-0) = 1$; $C_{23} = -(3-0) = -3$
- $C_{31} = +(0-2) = -2$; $C_{32} = -(0-4) = 4$; $C_{33} = +(1-0) = 1$

Arrange these cofactors into a matrix C , then transpose it to find the adjoint matrix ($\text{adj}(M) = C^T$):

$$\text{adj}(M) = \begin{vmatrix} 1 & 6 & -2 \\ -2 & 1 & 4 \\ 6 & -3 & 1 \end{vmatrix}$$

PROBLEM 21: DETERMINANT OF AN ADJOINT MATRIX SYSTEM

If A is a 3x3 square matrix with a determinant of $\det(A) = 5$, calculate the determinant of its adjoint matrix, $\det(\text{adj}(A))$.

SOLUTION:

Use the adjoint determinant identity: $\det(\text{adj}(A)) = (\det(A))^{n-1}$.

Given the order $n = 3$ and the determinant value $\det(A) = 5$:

$$\det(\text{adj}(A)) = 5^{3-1} = 5^2 = 25$$

Section 6: Inverse Matrix Operations

PROBLEM 22: INVERTING A 2X2 MATRIX

Calculate the inverse matrix A^{-1} for:

$$A = \begin{vmatrix} 3 & 2 \\ 4 & 3 \end{vmatrix}$$

SOLUTION:

Step 1: Calculate the determinant.

$|A| = (3 \times 3) - (2 \times 4) = 9 - 8 = 1$. Since the determinant is non-zero, the inverse exists.

Step 2: Find the adjoint matrix.

$adj(A) =$

$$\begin{vmatrix} 3 & -2 \\ -4 & 3 \end{vmatrix}$$

Step 3: Apply the inverse formula. Since $|A| = 1$, the inverse matrix equals the adjoint matrix:

$$A^{-1} = \begin{vmatrix} 3 & -2 \\ -4 & 3 \end{vmatrix}$$

PROBLEM 23: INVERTING A 2X2 MATRIX WITH A FRACTIONAL SCALE FACTOR

Find the inverse matrix B^{-1} for:

$$B = \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix}$$

SOLUTION:

Step 1: Calculate the determinant.

$$|B| = (2 \times 3) - (4 \times 1) = 6 - 4 = 2.$$

Step 2: Find the adjoint matrix.

$$\text{adj}(B) =$$

$$\begin{vmatrix} 3 & -4 \\ -1 & 2 \end{vmatrix}$$

Step 3: Divide by the determinant. Apply the formula $B^{-1} = (1 / |B|) \times \text{adj}(B)$:

$$B^{-1} = \begin{vmatrix} 1.5 & -2.0 \\ -0.5 & 1.0 \end{vmatrix}$$

PROBLEM 24: VERIFYING SINGULAR NON-INVERTIBILITY STATE

Show that the following matrix S cannot be inverted:

$$S = \begin{vmatrix} 3 & 6 \\ 2 & 4 \end{vmatrix}$$

SOLUTION:

A matrix is singular and cannot be inverted if its determinant is exactly zero.

Calculate the determinant of matrix S :

$$\det(S) = (3 \times 4) - (6 \times 2) = 12 - 12 = 0$$

Because $\det(S) = 0$, matrix S is singular, meaning its inverse S^{-1} is undefined.

PROBLEM 25: FULL 3X3 MATRIX INVERSE VERIFICATION

Find the inverse matrix M^{-1} for the matrix analyzed in Problem 20:

$$M = \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{vmatrix}$$

SOLUTION:

Step 1: Calculate the determinant of matrix M .

Expand along the first row:

$$|M| = 1 \times ((1 \times 1) - (0 \times 3)) - 0 + 2 \times ((2 \times 3) - (1 \times 0)) = 1(1) + 2(6) = 1 + 12 = 13$$

Since $|M| = 13 \neq 0$, the inverse exists.

Step 2: Retrieve the adjoint matrix from Problem 20.

$\text{adj}(M) =$

$$\begin{vmatrix} 1 & 6 & -2 \\ -2 & 1 & 4 \\ 6 & -3 & 1 \end{vmatrix}$$

Step 3: Apply the inverse formula $M^{-1} = (1 / |M|) * \text{adj}(M)$. Divide every entry in the adjoint matrix by 13:

$$M^{-1} = \begin{vmatrix} 1/13 & 6/13 & -2/13 \\ -2/13 & 1/13 & 4/13 \\ 6/13 & -3/13 & 1/13 \end{vmatrix}$$